

Experiment 'p_13_2_recursive_square_area' Results

June 5, 2026

Experiment outcome: SUCCESS

Bad responses: 0

Responses containing assume : 0

Responses containing expect : 0

Resolution attempts: 1

Hard fails (resolution): 0

Soft fails (resolution): 0

Verification attempts: 1

Problem Specification

Problem name: p_13_2_recursive_square_area

Natural language statement: null

Method signature: p_13_2_recursive_square_area(width: nat) returns (area: nat)

Ensures

- `area == width * width`

Requires

- `1 <= width`

GenAI interactions

Below you will find all interactions between the ‘user’ (program) and the ‘assistant’ (OpenAI).

Program → GenAI

You are given the following task to perform in Dafny:

Write a method to satisfy the below requirements.

The signature should be:

`p_13_2_recursive_square_area(width: nat) returns (area: nat)`

The method should respect the following contract:

`requires 1 <= width, ensures area == width * width`

Produce and show only the Dafny body of this method, including the curly braces that
→ surround it. Do not show the signature nor contract. You must not use 'assume' nor
→ 'expect' anywhere in your code.

GenAI → Program

System fingerprint: null

ID: chatcmpl-DnOo4CEkZ4rGrPtousPTi4Nq7Wvrr

Model: gpt-5.5-2026-04-23

Created at: 1780665800

Finish reason: stop

Usage: 256 tokens in, and 1510 tokens out

```
{  
  area := width * width;  
}
```

Final Program

```
method p_13_2_recursive_square_area(width: nat) returns (area: nat)  
  requires 1 <= width  
  ensures area == width * width  
{  
  area := width * width;  
}
```

Total Token Usage

Input tokens: 256

Output tokens: 1510

Reasoning tokens: 1470

Sum of ‘total tokens’: 1766

Experiment Timings

Overall Experiment started at 1780665797441, ended at 1780665892979, lasting 95538ms (95.54 seconds)

Iteration #1 started at 1780665797444, ended at 1780665892972, lasting 95528ms (95.53 seconds)